Final-Project Datasheet

Motivation

- **For what purpose was the dataset created?**
  I pulled this dataset together to train and tweak a Bayesian Optimization loop for an assignment. The core goal was to navigate eight different "black-box" functions, figure out how the continuous inputs (ranging from 2D up to 8D) map to a single output score, and ultimately find the settings that yield the highest possible peak performance.
- **Who created the dataset and on behalf of which entity? Who funded it?**
  The engineering team behind the university's machine learning capstone course provided the initial baseline data arrays (`initial_inputs.npy` and `initial_outputs.npy`). From there, I expanded the dataset week by week over thirteen weeks by generating new coordinates with my code and submitting them to the course server to collect the resulting evaluation scores. The entire project was hosted and funded by the university's academic department.

Composition

- **What do the instances that comprise the dataset represent?**
  Each entry is a pairing of an input coordinate vector (X) and its corresponding output score (Y). Each function space represents a different simulated engineering problem:
  - *Function 1:* Finding contamination points in a 2D radiation field.
  - *Functions 2, 7, and 8:* Tuning hyperparameter surfaces for unknown machine learning models.
  - *Function 3:* Balancing mixtures for a drug discovery simulator.
  - *Function 4:* Arranging layouts for a warehouse placement strategy.
  - *Function 5:* Maximizing the chemical yield of an industrial process plant.
  - *Function 6:* Optimizing a cake recipe parameter sweep.
- **How many instances of each type are there?**
  The dataset grew every single week as I added new evaluation queries. Here is how the total number of rows changed from the starting baseline files to my final Week 13 state:
  - **Functions 1 & 2 (2D):** Started with 10 baseline points → ended with 23 rows total.
  - **Function 3 (3D):** Started with 15 baseline points → ended with 28 rows total.

    - **Function 4 (4D):** Started with 30 baseline points → ended with 43 rows total.
    - **Function 5 (4D):** Started with 20 baseline points → ended with 33 rows total.
    - **Function 6 (5D):** Started with 20 baseline points → ended with 33 rows total.
    - **Function 7 (6D):** Started with 30 baseline points → ended with 43 rows total.
    - **Function 8 (8D):** Started with 40 baseline points → ended with 53 rows total.
- **Is there any missing data?**
  No, there are no empty slots or null values. Every single coordinate vector I submitted returned a clean, complete output score from the server, so the rows are fully populated.
- **Does the dataset contain data that might be considered confidential?**
  Not at all. The entire dataset consists of anonymized, synthetic numbers generated by simulators. There are no names, personal identifiers, or sensitive communications involved.

Collection Process

- **How was the data acquired?**
  I got the initial data arrays directly from the course page. For the weekly additions, I ran my custom Gaussian Process model to predict the highest peak using an Expected Improvement (EI) loop. My script would suggest a coordinate vector, I would query the server API, and then I appended the returning score straight back into my history arrays.
- **If the data is a sample of a larger subset, what was the sampling strategy?**
  The dataset is highly selective. Instead of scanning the spaces uniformly, my sampling strategy used a multi-start L-BFGS-B search engine (50 random restarts) to balance exploration with heavy exploitation. The code intentionally ignored low-performing areas and heavily focused on clusters surrounding the highest discovered peaks.
- **Over what time frame was the data collected?**
  The initial baseline metrics were packaged at the start of the semester, and I collected the sequential additions over a continuous 13-week course timeline.

Preprocessing/Cleaning/Labeling

- **Was any preprocessing/cleaning/labeling of the data done?**
  Yes, and it was a critical step. The output scales (Y) across these eight functions were completely mismatched—Function 1 yielded tiny fractions near

10^{-124}, while Function 5 shot past 1200. Feeding these raw scales directly into a standard Gaussian Process would completely break the hyperparameter tuning. To fix this, I forced a standard normal scaler layer (`normalize_y=True`) directly inside my regression loop. Every time the model fits the data, it centers the scores to a mean of zero and standardises them to unit variance.

- **Was the "raw" data saved in addition to the preprocessed data?**
  Yes. I kept all the original, unscaled `.npy` files intact in the root directories of this repository so that anyone can pull down the exact unedited numbers and experiment with different scaling approaches.

Uses

- **What other tasks could the dataset be used for?**
  This dataset is ideal for anyone looking to run comparative benchmarks on derivative-free global optimization frameworks. You can use it to test how alternative open-source frameworks (like `Optuna` or random forest models) stack up against a standard Gaussian Process loop on irregular, multi-dimensional surfaces.
- **Is there anything about the dataset that might impact future uses or create risks?**
  Yes, there is a major structural catch. Because this data was gathered sequentially by an active optimization algorithm, the points are **massively clustered** around the peaks. The dataset is riddled with selection bias. If a consumer tries to train a standard machine learning model (like a basic neural network or linear regression) on this dataset, it will struggle significantly to generalise, and it will have a massive blind spot regarding the low-performing regions. Anyone using this data needs to account for that spatial skew.
- **Are there tasks for which the dataset should not be used?**
  It should not be used to train models intended for global landscape visualisation or smooth space interpolation, simply because there are massive gaps in the coordinates where my code purposefully refused to look.

Distribution

- **How has the dataset already been distributed?**
  The foundational files were distributed internally through the university's course management portal. My sequential additions and final combinations are shared openly right here in this student version control repository.
- **Is it subject to any copyright or intellectual property (IP) license?**
  The underlying baseline arrays are owned by the university instructional team and are protected by standard academic integrity and non-distribution

guidelines. The 13 weeks of optimization updates I generated are shared under an open educational use license.

Maintenance

- **Who maintains the dataset?**
  I am actively maintaining, version-controlling, and archiving this repository history as the sole developer of this capstone submission.